



Time Series Analysis

Seminar 8: Long memory and ARFIMA



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Acronyms used in this material

- ▣ **ACF**: Autocorrelation Function
- ▣ **AI**: Artificial Intelligence
- ▣ **AR**: AutoRegressive (model)
- ▣ **ARCH**: AutoRegressive Conditional Heteroskedasticity
- ▣ **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- ▣ **ARIMA**: AutoRegressive Integrated Moving Average (model)
- ▣ **ARMA**: AutoRegressive Moving Average (model)
- ▣ **BET**: Bucharest Exchange Trading (index)
- ▣ **BIC**: Bayesian Information Criterion
- ▣ **BNR**: Banca Națională a României (the National Bank of Romania)
- ▣ **CPI**: Consumer Price Index
- ▣ **DFA**: Detrended Fluctuation Analysis
- ▣ **EODHD**: EOD Historical Data (market-data provider)
- ▣ **FIGARCH**: Fractionally Integrated GARCH
- ▣ **FRED**: Federal Reserve Economic Data
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GPH**: Geweke–Porter–Hudak (estimator)
- ▣ **HICP**: Harmonised Index of Consumer Prices
- ▣ **LW**: Local Whittle (estimator)
- ▣ **MA**: Moving Average
- ▣ **ML**: Machine Learning
- ▣ **OLS**: Ordinary Least Squares
- ▣ **RV**: Realised Volatility
- ▣ **SE**: Standard Error
- ▣ **US**: United States

Today's question and route

- **Question:** does a shock to a series fade at an exponential rate (ARMA) or much more slowly (long memory), and how do we measure the difference?
 - ▶ this seminar comes **before** Lecture 8: the section “Prerequisites for Today” gives every definition the tasks use
- Route
 - ▶ Part A: fractional weights and ARFIMA autocorrelations; R/S and GPH by hand; an ARFIMA forecast; GARCH and FIGARCH weights
 - ▶ Part B: the Nile and its break; Romanian inflation; S&P 500 volatility; BET and EUR/RON
 - ▶ Part C: US inflation and monetary regimes; an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	weights of $(1 - L)^d$, ACF of ARFIMA(0, d , 0), ranges of d	Solved, Proposed	A1
A3, A4	R/S and H by hand; a GPH slope by hand	Solved, Proposed	A3
A5, A6	an ARFIMA forecast; GARCH and FIGARCH weights	Solved, Proposed	A5
B1, B2	the Nile and its break; Romanian inflation	Solved, Proposed	B1
B3, B4	S&P 500 volatility; BET and EUR/RON	Solved, Proposed	B3
C1, C2	US inflation and regimes; what is wrong in an AI answer?	Proposed	B1, B2

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
Nile flow at Aswan	statsmodels (nile)	yearly	1871–1970
HICP, Romania	Eurostat (prc_hicp_minr)	monthly, 2015 = 100	2005–2026
CPI, United States	FRED (CPIAUCSL)	monthly, SA	1947–2026
S&P 500, BET	EODHD	daily, close	2000–2026
EUR/RON	BNR reference rate	daily	2005–2026

- Romanian inflation: monthly rate $100\Delta \ln P_t$ minus its monthly means (seasonal adjustment), and the 12-month rate $100(\ln P_t - \ln P_{t-12})$
- In the notebook: `gph`, `local_whittle`, `hurst_rs`, `arfima_exact_ml`, `monthly_rv`; no account or key is needed

Prerequisites for Today (1/22): autocorrelation and memory

- ▣ **Autocovariance** and **autocorrelation** of a stationary series x_t at lag k

$$\gamma(k) = \text{Cov}(x_t, x_{t-k}), \quad \rho(k) = \frac{\gamma(k)}{\gamma(0)}$$

- ▶ $\gamma(0) = \text{Var}(x_t)$; $\rho(k) \in [-1, 1]$: the correlation between two values k periods apart

- ▣ **Sample autocorrelation** (the ACF of the data, [Chapter 1](#))

$$\hat{\rho}(k) = \frac{\sum_{t=k+1}^T (x_t - \bar{x})(x_{t-k} - \bar{x})}{\sum_{t=1}^T (x_t - \bar{x})^2}$$

- ▶ T : the number of observations; $\bar{x}$: the sample mean; band of no correlation: $\pm 1.96/\sqrt{T}$

- ▣ **Memory**: how fast $\rho(k)$ goes to zero as the lag k grows

- ▶ fast decay: a shock is forgotten quickly; slow decay: the distant past still matters

Prerequisites for Today (2/22): short memory and long memory

- ▣ **Short memory** (ARMA models, [Chapter 2](#)): exponential decay

$$|\rho(k)| \leq C r^k, \quad 0 < r < 1$$

- ▶ $C > 0$: a constant; r : the decay factor per lag; AR(1): $\rho(k) = \phi^k$, so $r = |\phi|$
- ▶ the autocorrelations are summable: $\sum_k |\rho(k)| < \infty$

- ▣ **Long memory**: hyperbolic (power-law) decay

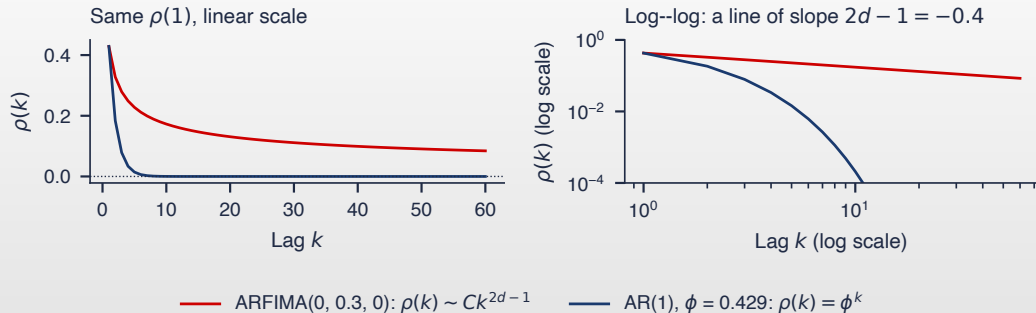
$$\rho(k) \sim C k^{2d-1}, \quad 0 < d < \frac{1}{2}$$

- ▶ d : the **memory parameter**; $\sim$: the ratio of the two sides tends to 1 as $k \rightarrow \infty$
- ▶ the exponent $2d - 1$ lies in $(-1, 0)$: the decay is so slow that $\sum_k \rho(k) = \infty$

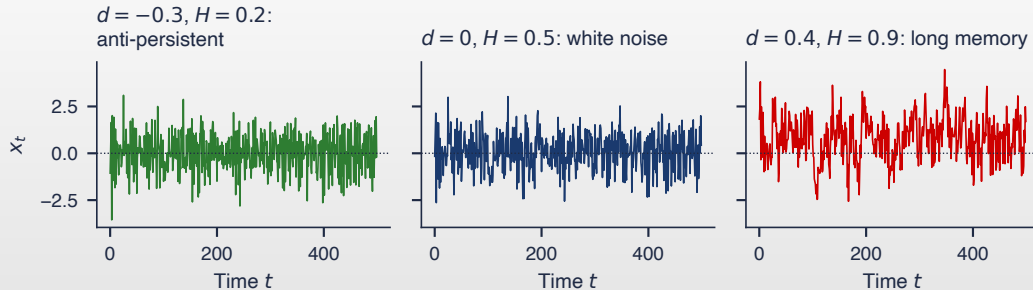
- ▣ **Hurst exponent** $H = d + 1/2$

- ▶ $H = 0.5$: no memory; $H > 0.5$: persistence; $H < 0.5$: anti-persistence
- ▶ Hurst found $H \approx 0.7$ for the yearly flow of the Nile [Hurst \(1951\)](#)

Prerequisites for Today (3/22): the two decays in a chart



- Left: ARFIMA(0, 0.3, 0) and an AR(1) with the same $\rho(1) = 0.429$
 - ▶ at lag 10: $\rho(10) = 0.173$ for ARFIMA, but only $0.429^{10} = 0.0002$ for the AR(1)
- Right: the same curves on log-log axes
 - ▶ $\log \rho(k) \approx \log C + (2d - 1) \log k$: a power law is a straight line of slope $2d - 1 = -0.4$
 - ▶ the exponential decay bends down and leaves the chart after about 10 lags

Prerequisites for Today (4/22): simulated paths for three values of d 

- The same shocks ε_t (white noise, variance 1), filtered with three values of d (the model is defined on the next slides)
- $d = -0.3$ ($H = 0.2$): **anti-persistent**
 - ▶ a value above the mean tends to be followed by one below: $\rho(1) < 0$, the path looks rougher than white noise
- $d = 0.4$ ($H = 0.9$): **persistent**
 - ▶ long excursions above or below the mean, which look like trends or changes of level, although the mean is constant

Prerequisites for Today (5/22): the lag operator and fractional differencing

- **Lag operator** L : $Lx_t = x_{t-1}$, $L^k x_t = x_{t-k}$
 - ▶ first difference: $(1 - L)x_t = x_t - x_{t-1}$, the case $d = 1$
- **Fractional difference** $(1 - L)^d$ for a real d : the binomial series

$$(1 - L)^d x_t = \sum_{k=0}^{\infty} \pi_k x_{t-k}, \quad \pi_0 = 1, \quad \pi_k = \pi_{k-1} \frac{k-1-d}{k}$$

- ▶ π_k : the weight of the value x_{t-k} ; $d = 1$ gives $\pi_1 = -1$ and $\pi_k = 0$ for $k \geq 2$ (the usual difference)
 - ▶ for $0 < d < 1$: every $\pi_k < 0$ ($k \geq 1$) and $|\pi_k|$ decreases slowly, like k^{-1-d} : all past values count
- Example, $d = 0.4$: $\pi_1 = -0.4$, $\pi_2 = -0.4 \cdot 0.6/2 = -0.12$, $\pi_3 = -0.12 \cdot 1.6/3 = -0.064$

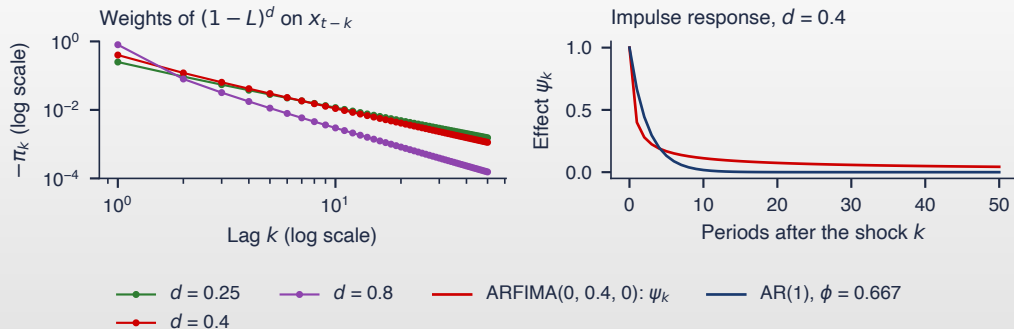
Prerequisites for Today (6/22): the impulse response ψ_k

- ▣ **Inverse filter:** the series as a sum of past shocks (the $MA(\infty)$ form)

$$x_t = (1 - L)^{-d} \varepsilon_t = \sum_{k=0}^{\infty} \psi_k \varepsilon_{t-k}, \quad \psi_0 = 1, \quad \psi_k = \psi_{k-1} \frac{k - 1 + d}{k}$$

- ▶ ψ_k : the effect on x_t of a unit shock that happened k periods ago
- ▣ Example, $d = 0.4$: $\psi_1 = 0.4$, $\psi_2 = 0.4 \cdot 1.4/2 = 0.28$, $\psi_3 = 0.28 \cdot 2.4/3 = 0.224$
 - ▶ for $0 < d < 1/2$, ψ_k decreases like k^{d-1} : slowly, but towards 0, so the series is stationary
- ▣ For an $AR(1)$, $\psi_k = \phi^k$: the effect of a shock dies out exponentially

Prerequisites for Today (7/22): fractional weights and impulse responses in a chart



- Left: the weights $-\pi_k$ of $(1 - L)^d$ on log-log axes are almost straight lines: a power law in k
- Right: for $d = 0.4$, $\psi_{10} = 0.112$, against $0.667^{10} = 0.017$ for an AR(1) with the same $\rho(1) = 0.667$

Prerequisites for Today (8/22): the ARFIMA model

- **ARFIMA**(p, d, q): an ARMA model for the fractionally differenced series [Granger and Joyeux \(1980\)](#), [Hosking \(1981\)](#)

$$\phi(L)(1-L)^d(x_t - \mu) = \theta(L)\varepsilon_t$$

- ▶ $\phi(L) = 1 - \phi_1 L - \dots - \phi_p L^p$, $\theta(L) = 1 + \theta_1 L + \dots + \theta_q L^q$: the AR and MA polynomials ([Chapter 2](#))
- ▶ μ : the mean; ε_t : white noise with variance σ^2 ; $d = 0$: ARMA; $d = 1$: ARIMA($p, 1, q$)

Range of d	Properties
$d \leq -1/2$	not invertible
$-1/2 < d < 0$	stationary, invertible, anti-persistent
$d = 0$	short memory (ARMA)
$0 < d < 1/2$	stationary, invertible, long memory
$1/2 \leq d < 1$	not stationary, but mean-reverting: a shock still dies out
$d = 1$	unit root: a shock is permanent

- For $1/2 \leq d < 3/2$, difference once: $(1-L)x_t$ is ARFIMA with memory parameter $d - 1$

Prerequisites for Today (9/22): ARFIMA(0, d , 0) in numbers

▣ **Autocorrelations** by recursion

$$\rho(1) = \frac{d}{1-d}, \quad \rho(k) = \rho(k-1) \frac{k-1+d}{k-d}, \quad k \geq 2$$

- ▶ example $d = 0.3$: $\rho(1) = 0.3/0.7 = 0.429$, $\rho(2) = 0.429 \cdot 1.3/1.7 = 0.328$
- ▶ for large k : $\rho(k) \approx \frac{\Gamma(1-d)}{\Gamma(d)} k^{2d-1}$, the constant C of the long-memory definition

▣ **Variance**

$$\gamma(0) = \sigma^2 \frac{\Gamma(1-2d)}{\Gamma(1-d)^2}$$

- ▶ $\Gamma(z) = \int_0^\infty u^{z-1} e^{-u} du$: the gamma function; $\Gamma(n) = (n-1)!$ for integers $n \geq 1$; $\Gamma(1/2) = \sqrt{\pi}$
- ▶ $d > 0$: $\gamma(0) > \sigma^2$, the memory accumulates past shocks; $\gamma(0) \rightarrow \infty$ as $d \rightarrow 1/2$

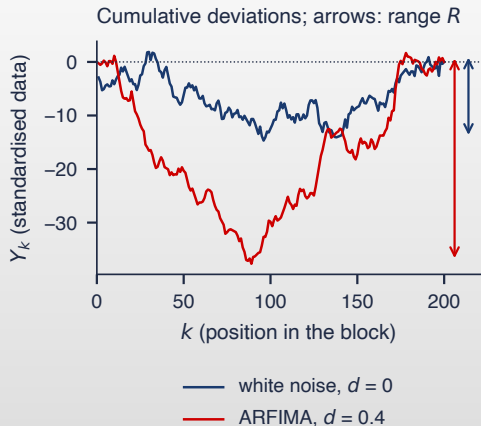
Prerequisites for Today (10/22): the one-step forecast

- From $(1 - L)^d(x_{T+1} - \mu) = \varepsilon_{T+1}$, replace the unknown shock ε_{T+1} by its mean, 0

$$\hat{x}_{T+1} = \mu - \sum_{k \geq 1} \pi_k (x_{T+1-k} - \mu)$$

- ▶ x_T : the last observation; x_{T+1-k} : the value $k - 1$ periods before it; in practice the sum stops at $k = T$
- Reading the formula for $0 < d < 1$
 - ▶ the weights $-\pi_k$ are positive: each past deviation from μ pushes the forecast in its own direction
 - ▶ the weights decrease slowly, so old deviations still count
- Comparison: AR(1), $\hat{x}_{T+1} = \mu + \phi(x_T - \mu)$
 - ▶ only the last value enters the forecast; ϕ : the AR(1) coefficient

Prerequisites for Today (11/22): the R/S statistic



- For a block $x_1, \dots, x_n$ with mean $\bar{x}$ [Hurst \(1951\)](#):

$$Y_k = \sum_{i=1}^k (x_i - \bar{x}), \quad k = 1, \dots, n$$

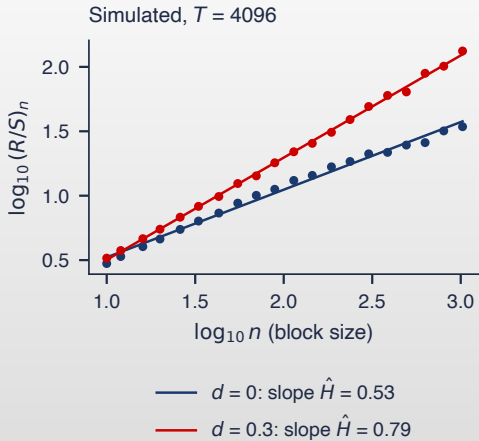
- ▶ Y_k : the cumulative deviation from the mean; $Y_n = 0$

- **Range and standard deviation**

$$R = \max_k Y_k - \min_k Y_k, \quad S = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

- ▶ R/S : the range of the cumulative deviations, in units of S ; it grows like n^H
- Chart: a persistent block has long runs of the same sign
 - ▶ Y_k drifts far from 0, so R is large (39 against 17 for white noise)

Prerequisites for Today (12/22): H from a log-log regression



- Split the series in blocks of size n and average R/S over the blocks: $(R/S)_n$

$$\log_{10}(R/S)_n \approx a + H \log_{10} n$$

- ▶ repeat for several n ; $\hat{H}$ = the OLS slope of $\log_{10}(R/S)_n$ on $\log_{10} n$; $\hat{d} = \hat{H} - 0.5$
- ▶ a : the intercept; OLS: ordinary least squares
- Chart: white noise gives $\hat{H} = 0.53$, slightly above 0.5
 - ▶ R/S is biased upwards in small blocks; $d = 0.3$ gives $\hat{H} = 0.79 \approx 0.8$
- DFA (detrended fluctuation analysis) [Peng et al. \(1994\)](#)
 - ▶ cumulate the series, fit a line in each block of size n ; $F(n)$: the root mean square of the residuals
 - ▶ $F(n) \propto n^H$: $\hat{H}$ = the slope of $\log F(n)$ on $\log n$

Prerequisites for Today (13/22): the periodogram and the spectral density

- **Fourier frequencies** $\lambda_j = 2\pi j/T$, $j = 1, \dots, \lfloor T/2 \rfloor$
 - ▶ λ_j in radians per period; it corresponds to a cycle of T/j periods; $\lfloor \cdot \rfloor$: the integer part
 - ▶ low frequencies (small j): long cycles; this is where long memory shows
- **Periodogram**: how much of the variance is in cycles of frequency λ_j

$$I(\lambda_j) = \frac{1}{2\pi T} \left| \sum_{t=1}^T (x_t - \bar{x}) e^{-i\lambda_j t} \right|^2$$

- ▶ i : the imaginary unit; $I(\lambda_j)$ is a noisy estimate of the spectral density $f(\lambda_j)$
- **Long memory** = a pole of the spectral density at frequency zero

$$f(\lambda) \sim G \lambda^{-2d} \quad \text{as } \lambda \rightarrow 0$$

- ▶ $G > 0$: a constant; ARFIMA(0, d , 0): $f(\lambda) = \frac{\sigma^2}{2\pi} (4 \sin^2(\lambda/2))^{-d}$, and $4 \sin^2(\lambda/2) \approx \lambda^2$ for small λ

Prerequisites for Today (14/22): the GPH estimator

- **GPH Geweke and Porter-Hudak (1983)**: take logs of $I(\lambda_j) \approx f(\lambda_j)$ at the m lowest frequencies: a linear regression in d

$$\log I(\lambda_j) = c + d X_j + u_j, \quad X_j = -\log(4 \sin^2(\lambda_j/2)), \quad j = 1, \dots, m$$

- ▶ c : the intercept; u_j : the error; m : the **bandwidth**, the number of low frequencies used

- The estimate is the OLS slope, with $y_j = \log I(\lambda_j)$

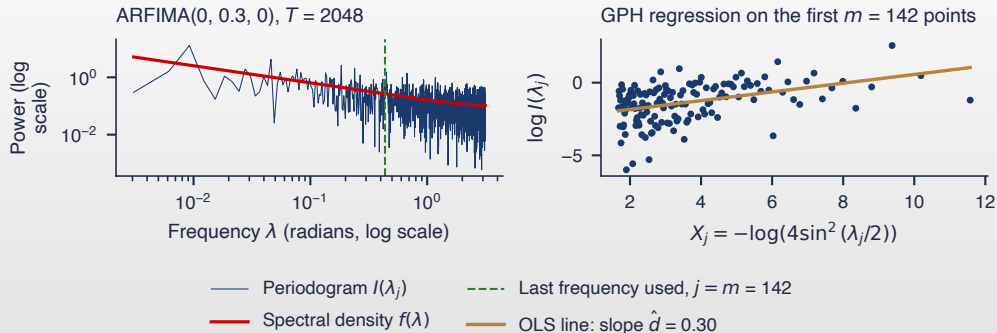
$$\hat{d} = \frac{\sum_{j=1}^m (X_j - \bar{X})(y_j - \bar{y})}{\sum_{j=1}^m (X_j - \bar{X})^2}, \quad \text{SE}(\hat{d}) = \frac{\pi}{\sqrt{24m}}$$

- ▶ $\bar{X}$, $\bar{y}$: the means over the m frequencies; SE: the asymptotic standard error

- **Test of H_0 : $d = 0$** (no long memory)

- ▶ $t = \hat{d}/\text{SE}(\hat{d})$, compared with the Normal distribution: reject at 5% if $|t| > 1.96$
- ▶ usual bandwidth $m = \lfloor T^{0.65} \rfloor$; a small m : less bias from short-run dynamics, but more noise

Prerequisites for Today (15/22): the periodogram and GPH in a chart



- Left: the periodogram of a simulated ARFIMA(0, 0.3, 0) ($T = 2048$) scatters around $f(\lambda)$, which rises towards $\lambda = 0$
- Right: the GPH regression on the first $m = \lfloor 2048^{0.65} \rfloor = 142$ frequencies gives $\hat{d} = 0.30$ ($\text{SE} = \pi / \sqrt{24 \cdot 142} = 0.054$)

Prerequisites for Today (16/22): local Whittle, exact ML and the information criteria

- **Local Whittle (LW)** [Robinson \(1995\)](#): $\hat{d}$ minimises, over the same m frequencies,

$$R(d) = \log \left(\frac{1}{m} \sum_{j=1}^m \lambda_j^{2d} I(\lambda_j) \right) - \frac{2d}{m} \sum_{j=1}^m \log \lambda_j$$

- ▶ a likelihood that uses only $f(\lambda) \approx G\lambda^{-2d}$ near zero; $\text{SE}(\hat{d}) = 1/(2\sqrt{m})$, smaller than for GPH

- **Exact maximum likelihood (ML)** [Sowell \(1992\)](#)

- ▶ maximises the full Normal likelihood L of ARFIMA(p, d, q) over d , the ϕ_i , the θ_j and σ^2
- ▶ uses all frequencies: more precise if the model is right, biased if the short-run part is wrong

- **Information criteria**: the model with the smallest value wins

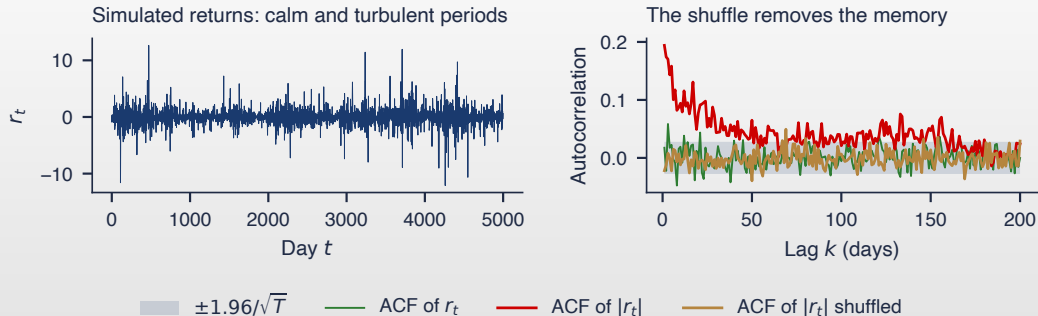
$$\text{AIC} = -2 \ln L + 2k, \quad \text{BIC} = -2 \ln L + k \ln T$$

- ▶ In L : the maximised log-likelihood; k : the number of estimated parameters; BIC penalises extra parameters more

Prerequisites for Today (17/22): memory in volatility and the shuffle test

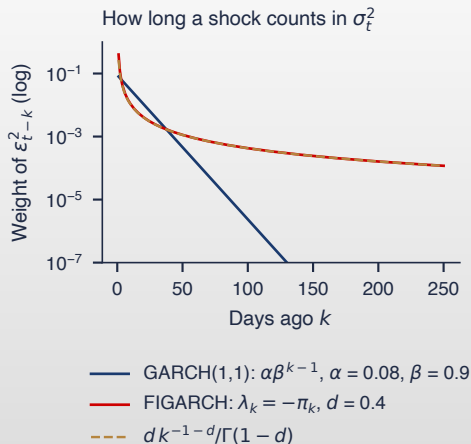
- Returns $r_t = 100 \ln(P_t/P_{t-1})$ are almost uncorrelated, but their size $|r_t|$ has long memory [Ding, Granger and Engle \(1993\)](#)
 - ▶ P_t : the closing price of day t ; monthly **realised volatility** $\sqrt{RV_m}$, with $RV_m = \sum_{t \in m} r_t^2$ (the days t of month m); we model $\log RV_m$
- **Shuffle test**: permute the days at random and re-estimate d
 - ▶ the permutation keeps the distribution of the values and destroys their order: real memory gives $\hat{d} \approx 0$ after shuffling
- Why: memory in $|r_t|$ means that calm and turbulent periods last (volatility clustering)
 - ▶ it concerns the size of the moves, not their sign: the direction stays unpredictable

Prerequisites for Today (18/22): the shuffle test in a chart



Simulated: ACF of $|r_t|$ positive up to lag 200, LW $\hat{d} = 0.28$; after shuffling, inside the band, $\hat{d} = -0.04$

Prerequisites for Today (19/22): GARCH and FIGARCH weights



- **GARCH(1,1)** (Chapter 5): $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$
 - ▶ σ_t^2 : the conditional variance of day t ; ε_t : the shock; $\omega > 0$, $\alpha, \beta \geq 0$: the parameters
 - ▶ substituting σ_{t-1}^2 , σ_{t-2}^2 , ... again and again (ARCH(∞) form):

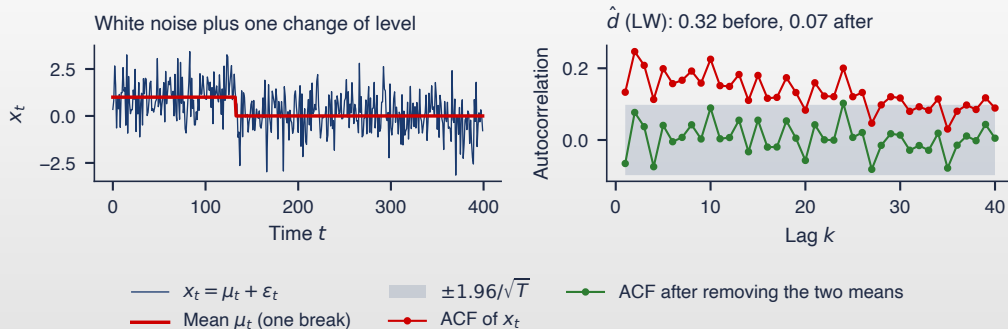
$$\sigma_t^2 = \frac{\omega}{1-\beta} + \alpha \sum_{k \geq 1} \beta^{k-1} \varepsilon_{t-k}^2$$

- ▶ weight of ε_{t-k}^2 : $\alpha\beta^{k-1}$, exponential
- **FIGARCH(0, d , 0)** Baillie, Bollerslev and Mikkelsen (1996): $\sigma_t^2 = \omega + [1 - (1-L)^d] \varepsilon_t^2$
 - ▶ weights $\lambda_k = -\pi_k \approx dk^{-1-d}/\Gamma(1-d)$: hyperbolic
 - ▶ chart, $k = 100$: $2.4 \cdot 10^{-6}$ (GARCH) against $4.3 \cdot 10^{-4}$ (FIGARCH)

Prerequisites for Today (20/22): spurious long memory

- A short-memory series with a change of level: $x_t = \mu_t + \varepsilon_t$, where the mean μ_t changes once (a **break**)
 - ▶ the ACF stays positive at long lags and $\hat{d} > 0$, although there is no memory [Diebold and Inoue \(2001\)](#)
 - ▶ remedy: estimate the mean of each regime, subtract it and re-estimate d
- **Monte Carlo band**: simulate many i.i.d. series of the same length, estimate d on each
 - ▶ the 2.5% and 97.5% quantiles of these $\hat{d}$: the 95% band of $\hat{d}$ when there is no memory; i.i.d.: independent and identically distributed
- An estimate $\hat{d}$ outside the band: evidence of memory, or of a break; inside: compatible with no memory

Prerequisites for Today (21/22): a break that looks like long memory



- Left: white noise whose mean falls by 1 after one third of the sample ($T = 400$)
- Right: the ACF stays above the band for 40 lags and LW gives $\hat{d} = 0.32$; after removing the two means, $\hat{d} = 0.07$

Prerequisites for Today (22/22): monthly and 12-month inflation rates

- P_t : the price index of month t ; **monthly rate** $\pi_t^m = 100 \Delta \ln P_t = 100(\ln P_t - \ln P_{t-1})$
 - ▶ seasonal adjustment by monthly means: subtract from each month (January, February, ...) its average over all years
- The **12-month rate** is the sum of the last 12 monthly rates

$$100(\ln P_t - \ln P_{t-12}) = \sum_{j=0}^{11} \pi_{t-j}^m$$

- ▶ two consecutive 12-month rates share 11 monthly rates (overlapping sums)
- ▶ even for i.i.d. monthly rates, the 12-month rate has $\rho(k) = (12 - k)/12$ for $k < 12$: a strong autocorrelation built by the definition

A1: fractional weights and the ACF [Solved]

Task

- A series is ARFIMA(0, 0.25, 0) with $\sigma^2 = 1$.
- 1. Compute $\pi_1, \dots, \pi_4$ of $(1 - L)^{0.25}$.
- 2. Compute $\rho(1)$ and $\rho(2)$, and give H and the class of the process.
- 3. Compare $\rho(5)$ with that of an AR(1) with the same $\rho(1)$.
- Report: four weights, two autocorrelations, H , two values of $\rho(5)$.

Solution

- 1. $\pi_1 = -0.25$; $\pi_2 = -0.25 \cdot 0.75/2 = -0.0938$;
 $\pi_3 = -0.0938 \cdot 1.75/3 = -0.0547$;
 $\pi_4 = -0.0547 \cdot 2.75/4 = -0.0376$
- 2. $\rho(1) = 0.25/0.75 = 0.333$;
 $\rho(2) = 0.333 \cdot 1.25/1.75 = 0.238$; $H = 0.75$: stationary, long memory
- 3. ARFIMA: $\rho(5) = 0.151$; AR(1): $0.333^5 = 0.0041$, about 37 times smaller

A2: which range is d in? [Proposed]

Solution discussed in class

Task

- ▣ Two series: (a) ARFIMA(0, 0.6, 0); (b) ARFIMA(0, -0.3, 0). Model: A1.
- ▣ 1. Compute π_1 and π_2 of $(1 - L)^d$ for each.
- ▣ 2. Say whether each series is stationary, invertible and mean-reverting.
- ▣ 3. For (b) compute $\rho(1)$ and $\rho(2)$; for (a) compute $\rho(1)$ of the first difference.
- ▣ Report: four weights, two classifications, three autocorrelations.

A3: R/S by hand and H from three points [Solved]

Task

- ▣ Block: $x = (0.6, 0.4, 0.5, 0.3, -0.4, -0.6, -0.2, -0.5)$. An R/S plot gives $(R/S)_n = 4.1; 9.6; 22.4$ for $n = 16, 64, 256$.
- ▣ 1. Compute $\bar{x}$, the cumulative deviations Y_k , R , S and R/S .
- ▣ 2. Compute H as the slope of $\log_{10}(R/S)_n$ on $\log_{10} n$, and $d = H - 0.5$.
- ▣ Report: R/S , H , d and one sentence.

Solution

- ▣ 1. $\bar{x} = 0.0125$; Y_k : 0.5875, 0.9750, 1.4625, 1.7500, 1.3375, 0.7250, 0.5125, 0; $R = 1.750$, $S = 0.457$, $R/S = 3.83$
- ▣ the signs are grouped (four positive, then four negative): the range is large, as for a persistent series
- ▣ 2. $\log_{10} n$: 1.204, 1.806, 2.408; $\log_{10}(R/S)$: 0.613, 0.982, 1.350; slope $H = 0.61$, $d = 0.11$: moderate persistence

A4: a GPH slope by hand [Proposed]

Solution discussed in class

Task

- ▣ For $m = 4$ frequencies: $X_j = 0.5; 1.6; 2.3; 3.0$ and $\log I(\lambda_j) = 1.2; 1.8; 2.0; 2.5$. Model: A3.
- ▣ 1. Compute $\bar{X}$, $\bar{y}$, $\sum(X_j - \bar{X})(y_j - \bar{y})$ and $\sum(X_j - \bar{X})^2$.
- ▣ 2. Compute $\hat{d}$ and its standard error $\pi/\sqrt{24m}$.
- ▣ 3. Test $H_0: d = 0$ at 5% and comment on the size of m .
- ▣ Report: four sums, $\hat{d}$, its SE, a decision and one sentence.



A5: a one-step ARFIMA forecast [Solved]

Task

- ▣ ARFIMA(0, 0.4, 0) with $\mu = 2$; the last four values: $x_T = 3.0$, $x_{T-1} = 2.5$, $x_{T-2} = 2.8$, $x_{T-3} = 1.5$.
- ▣ 1. Write the first four weights $-\pi_k$ of $(1 - L)^{0.4}$.
- ▣ 2. Compute $\hat{x}_{T+1}$ using these four lags.
- ▣ 3. Compare with an AR(1) with the same $\rho(1)$.
- ▣ Report: four weights, two forecasts and one sentence.

Solution

- ▣ 1. $-\pi_k$: 0.4; 0.12; 0.064; 0.0416
- ▣ 2. $\hat{x}_{T+1} =$
 $2 + 0.4(1.0) + 0.12(0.5) + 0.064(0.8) + 0.0416(-0.5) =$
 $2 + 0.4000 + 0.0600 + 0.0512 - 0.0208 = 2.490$
- ▣ 3. AR(1): $\phi = \rho(1) = 0.4/0.6 = 0.667$,
 $\hat{x}_{T+1} = 2 + 0.667 \cdot 1.0 = 2.667$; the AR(1) looks only at x_T , ARFIMA spreads the weight over the past (in practice over all T values)

A6: how long does a volatility shock count? [Proposed]

Solution discussed in class

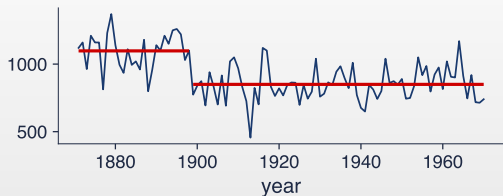
Task

- ▣ GARCH(1,1) with $\alpha = 0.10$, $\beta = 0.88$; FIGARCH(0, d , 0) with $d = 0.4$ (weights $\lambda_k = -\pi_k$). Model: A5.
- ▣ 1. Compute the weight of ε_{t-k}^2 in σ_t^2 for $k = 1, 2, 3$ in both models.
- ▣ 2. Compute both weights at $k = 20$ and $k = 100$; for FIGARCH use $\lambda_k \approx d k^{-1-d} / \Gamma(1-d)$, $\Gamma(0.6) = 1.489$.
- ▣ 3. Explain which model keeps a crisis in today's volatility longer.
- ▣ Report: ten weights and one sentence.

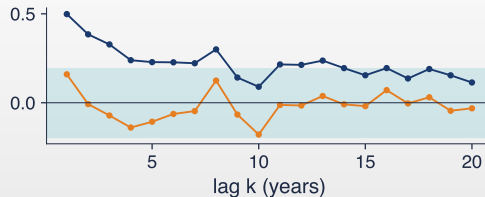
B1: does the Nile have long memory? [Solved]

- ▣ **Question:** is the persistence of the Nile flow long memory, or the effect of the change in 1898?
- ▣ **Inputs:** yearly flow at Aswan, 1871–1970 ($T = 100$)
- ▣ **Tasks**
 1. Plot the series and its ACF up to lag 20.
 2. Estimate H by R/S, d by GPH and local Whittle ($m = \lfloor T^{0.65} \rfloor$) and by exact ML of ARFIMA(0, d , 0).
 3. Subtract the mean of 1871–1898 and the mean of 1899–1970 and repeat step 2.
 4. Interpretation: does the Nile have long memory?
- ▣ **Report:** two ACF values, four estimates before and after, one sentence
- ▣ **Notebook:** section B1

B1: solution [Solved]



— Nile flow at Aswan (10^8 m^3)
 — regime means, break after 1898
 — ACF, raw series



— ACF, regime means removed
 95% band for white noise

- Raw: $\hat{\rho}(1) = 0.50$, $\hat{\rho}(10) = 0.09$; $H_{R/S} = 0.78$; GPH 0.44 (SE 0.15), LW 0.40 (SE 0.11), ML 0.36 (SE 0.07); $m = 19$
- Means 1098 and 850; after removing them: $\hat{\rho}(1) = 0.16$; $H_{R/S} = 0.63$; GPH -0.07 , LW -0.17 , ML 0.05 (SE 0.10)
- Interpretation: for 1871–1970 the evidence of long memory comes from one break; R/S stays above 0.5 because it is biased upwards in short samples

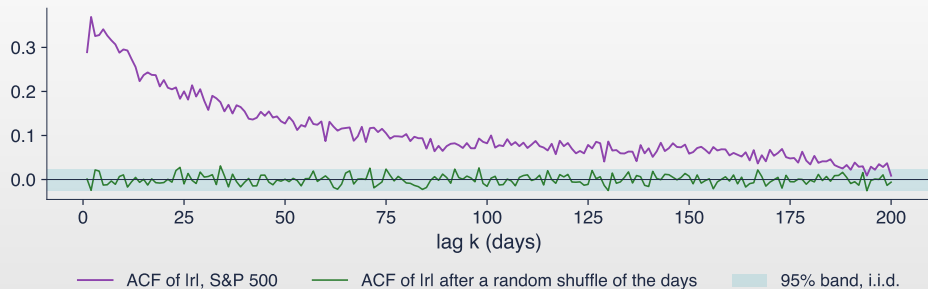
B2: long memory in Romanian inflation [Proposed]

- **Question:** how persistent is Romanian inflation, and does the way we measure inflation change the answer?
- **Inputs:** monthly HICP, 2005–2026: the monthly rate seasonally adjusted with the monthly means, and the 12-month rate; model: B1
- **Tasks**
 1. Estimate d of the monthly rate by GPH and local Whittle with $m = \lfloor T^a \rfloor$, $a = 0.5; 0.65; 0.8$.
 2. Fit ARFIMA(0, d , 0) and ARMA(1,1) by exact ML and compare their BIC.
 3. Estimate d of the 12-month rate by local Whittle and compare the two ACFs.
 4. Interpretation: why does the 12-month rate give d close to 1?
- **Report:** six estimates, two BIC values, one d , two sentences
- **Notebook:** section B2

B3: memory in S&P 500 volatility [Solved]

- **Question:** where is the memory of stock returns: in the returns or in their size?
- **Inputs:** daily log returns of the S&P 500 since 2000 and their monthly realised volatility
- **Tasks**
 1. Estimate d of r_t and of $|r_t|$ by local Whittle and GPH.
 2. Shuffle the days at random and re-estimate d of $|r_t|$; plot both ACFs of $|r_t|$.
 3. Estimate d of the monthly log realised volatility.
 4. Interpretation: what does the shuffle test show?
- **Report:** eight estimates, the chart and one sentence
- **Notebook:** section B3

B3: solution [Solved]



- $T = 6714$ days; r_t : LW -0.01 , GPH 0.02 (SE 0.03 , 0.04); $|r_t|$: LW 0.53 , GPH 0.56
- Shuffled $|r_t|$: LW -0.03 , GPH -0.03 ; ACF of $|r_t|$: 0.289 (lag 1), 0.127 (lag 50), 0.009 (lag 200); log RV (321 months): LW 0.41 , GPH 0.37
- Interpretation: the shuffle keeps every daily value and destroys their order; the memory disappears, so it lives in the clustering of calm and turbulent periods, not in the distribution of returns

B4: BET and EUR/RON [Proposed]

- **Question:** do the BET and the EUR/RON rate show the same memory, and is it stable before and after the 2008 crisis?
- **Inputs:** daily log returns of the BET (since 2000) and of the EUR/RON reference rate (since July 2005); model: B3
- **Tasks**
 1. Estimate d of r_t and $|r_t|$ by local Whittle on the full sample.
 2. Repeat on the subsamples up to 2007 and 2010–2026.
 3. Compare the estimates for r_t with the 95% Monte Carlo band of i.i.d. series of 1900 days.
 4. Interpretation: is the full-sample memory of BET returns real?
- **Report:** twelve estimates, one band and two sentences
- **Notebook:** section B4

C1: US inflation and monetary regimes [Proposed]

- **Question:** is the long memory of US inflation a property of inflation, or of the changes in monetary policy?
- **Inputs:** US monthly CPI inflation, annualised, 1947–2026; regimes 1947–1984, 1985–2019, 2020–2026; models: B1, B2
- **Tasks**
 1. Estimate d by local Whittle on the full sample and in each regime.
 2. Remove the three regime means and re-estimate d .
 3. Propose one more check (a break test, a Markov-switching model, a forecast comparison).
 4. Interpretation: can a constant d describe 80 years of inflation?
- **Report:** five estimates, one check and a project plan
- **Notebook:** section C1

C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant about the long memory of Romanian inflation and of the S&P 500. The answer:
- ▣ (a) Monthly inflation has $d = 0.31$, so its Hurst exponent is $H = d - 0.5 < 0$: it is anti-persistent.
- ▣ (b) The 12-month rate gives $d = 1.00$, so Romanian inflation has a unit root.
- ▣ (c) With $d = 0.31$ the series is stationary and its shocks die out.
- ▣ (d) Use `statsmodels.tsa.arima.model.ARIMA(x, order=(0, 0.31, 0))` to fit the ARFIMA model.
- ▣ (e) S&P 500 returns have $d = -0.01$ and $|r|$ has $d = 0.53$, so returns are predictable.
- ▣ (f) A significant GPH estimate proves long memory.
- ▣ **Tasks**
 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
- ▣ **Report:** a list of six verdicts with one line of justification each

Key takeaways

- ▣ $(1 - L)^d$: weights by the recursion $\pi_k = \pi_{k-1}(k - 1 - d)/k$; ARFIMA(0, d , 0): $\rho(1) = d/(1 - d)$
- ▣ $H = d + 1/2$; R/S and GPH are slopes of log-log regressions
- ▣ Romanian monthly inflation: $d \approx 0.3$; S&P 500: memory in $|r_t|$, not in r_t
- ▣ Breaks can create long memory: the Nile, BET returns, US inflation
- ▣ An AI answer is a draft: check H against d , the library, and the data transformation

After the seminar

- Lecture 8 develops each topic of today
 - ▶ hyperbolic decay, fractional differencing, ARFIMA, the estimators, forecasting, volatility, spurious memory
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: inflation memory and monetary regimes in Romania
- Reading: [Baillie \(1996\)](#); [Graves et al. \(2017\)](#); [Ding, Granger and Engle \(1993\)](#)
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

References (1/2)

- ▣ Baillie, R. T. (1996). Long memory processes and fractional integration in econometrics. *Journal of Econometrics*, 73(1), 5–59. doi:10.1016/0304-4076(95)01732-1
- ▣ Baillie, R. T., Bollerslev, T., & Mikkelsen, H. O. (1996). Fractionally integrated generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 74(1), 3–30. doi:10.1016/S0304-4076(95)01749-6
- ▣ Diebold, F. X., & Inoue, A. (2001). Long memory and regime switching. *Journal of Econometrics*, 105(1), 131–159. doi:10.1016/S0304-4076(01)00073-2
- ▣ Ding, Z., Granger, C. W. J., & Engle, R. F. (1993). A long memory property of stock market returns and a new model. *Journal of Empirical Finance*, 1(1), 83–106. doi:10.1016/0927-5398(93)90006-D
- ▣ Geweke, J., & Porter-Hudak, S. (1983). The estimation and application of long memory time series models. *Journal of Time Series Analysis*, 4(4), 221–238. doi:10.1111/j.1467-9892.1983.tb00371.x
- ▣ Granger, C. W. J., & Joyeux, R. (1980). An introduction to long-memory time series models and fractional differencing. *Journal of Time Series Analysis*, 1(1), 15–29. doi:10.1111/j.1467-9892.1980.tb00297.x

References (2/2)

- ▣ Graves, T., Gramacy, R., Watkins, N., & Franzke, C. (2017). A brief history of long memory: Hurst, Mandelbrot and the road to ARFIMA, 1951–1980. *Entropy*, 19(9), 437. doi:10.3390/e19090437
- ▣ Hosking, J. R. M. (1981). Fractional differencing. *Biometrika*, 68(1), 165–176. doi:10.1093/biomet/68.1.165
- ▣ Hurst, H. E. (1951). Long-term storage capacity of reservoirs. *Transactions of the American Society of Civil Engineers*, 116(1), 770–799. doi:10.1061/TACEAT.0006518
- ▣ Peng, C.-K., Buldyrev, S. V., Havlin, S., Simons, M., Stanley, H. E., & Goldberger, A. L. (1994). Mosaic organization of DNA nucleotides. *Physical Review E*, 49(2), 1685–1689. doi:10.1103/PhysRevE.49.1685
- ▣ Robinson, P. M. (1995). Gaussian semiparametric estimation of long range dependence. *The Annals of Statistics*, 23(5), 1630–1661. doi:10.1214/aos/1176324317
- ▣ Sowell, F. (1992). Maximum likelihood estimation of stationary univariate fractionally integrated time series models. *Journal of Econometrics*, 53(1–3), 165–188. doi:10.1016/0304-4076(92)90084-5